

	Solr	Elasticsearch
Installation and Configuration	Supported by detailed documentation	More intuitive
Indexing/Searching	Text-oriented	Better performance of analytical queries
Scalability and Clustering	Provides SolrCloud	Better Internet scalability and designed for the cloud
Community	A much bigger ecosystem of community	A growing community though not a complete open source mindset
Documentation	Very well-documented	Lacks in documentation

Figure 7: Differences between Solr and ES

You can query by clicking on the *query* tab which looks like what's shown in Figure 6.

Solr vs Elastic Search (ES)

Solr and Elastic Search are competing search servers. Both are built on top of Lucene, so their core features are identical. In fact, Solr and ES are so similar that an ES plug-in allows you to use Solr client/tools with ES. For most functions, there is no genuine cause to choose Solr over ES, or vice versa. But there are some minor differences listed in the table shown in Figure 7.

Deploying Solr with Tomcat

Please note, this deployment is done in Solr 5.1. The prerequisites are listed below.

1. Install the Tomcat Servlet Container on the system as follows:

```
C:\Solr with Tomcat\apache-
tomcat-7.0.57
```

(We will refer to it as *TOMCAT_HOME*.)

2. Download and extract the Solr package as follows:

```
C:\Solr-5.1.0\Solr-5.1.0
```

Now follow the steps given below.

1. Copy the Solr directory from the extracted package (C:\Solr-5.1.0\Solr-5.1.0\server\Solr) into the home directory of Tomcat—into *TOMCAT_HOME\bin* or into *TOMCAT_HOME* if Tomcat is to be

started as a service.

- 2) Copy the Solr war file (C:\Solr-5.1.0\Solr-5.1.0\server\webapps\Solr.war) into the Tomcat Web apps directory, *TOMCAT_HOME\webapps*. Tomcat will automatically deploy it.
- 3) Copy all the five jar files from the Solr extracted package (C:\Solr-5.1.0\Solr-5.1.0\server\lib\ext) to the *TOMCAT_HOME\lib* directory.

```
* jul-to-slf4j-1.7.7,
* jcl-over-slf4j-1.7.7,
* slf4j-log4j12-1.7.7,
* slf4j-api-1.7.7,
* log4j-1.2.17
```

- 4) Copy the *log4j.properties* file from the C:\Solr-5.1.0\Solr-5.1.0\server\resources directory to the *TOMCAT_HOME\lib* directory.
- 5) After *Solr.war* is extracted:
 - a) Create the core (if not already done).
 - b) Copy the further needed listener-specific jars to C:\Solr with the *Tomcat\apache-tomcat-7.0.57\webapps\Solr\WEB-INF\lib* directory.
 - c) Configure the entry for listener-related jars to the scheduler in the C:\Solr With Tomcat\apache-tomcat-7.0.57\webapps\Solr\WEB-INF\web.xml file.
 - d) Modify the server port in

the properties file of the scheduler (for example, C:\Solr with *Tomcat\apache-tomcat-7.0.57\bin\Solr\conf\dataimportScheduler.properties*).

- e) In order to run the *data-importhandler*, copy the jars specific to the *data-importhandler* and for the custom transformer, if any, in the folder C:\Solr With *Tomcat\apache-tomcat-7.0.57\bin\Solr\lib*.

```
* Solr-dataimporthandler-5.1.0.jar
* Solr-dataimporthandler-extras-
5.1.0.jar
* ojdbc14-10.2.0.5.jar
* SqltoSolrDateTransformer.jar
and change the entry in Solrconfig.xml
for the same as

<!-- for data import handler -->
<lib dir="..\lib/" regex="Solr-
dataimporthandler-\d.*.jar" />
<lib dir="..\lib/" regex="ojdbc14-
\d.*.jar" />
<!-- For Custom Date Transformer -->
<lib dir="..\lib/"
regex="SqltoSolrDateTransformer.jar" />
```

Start the Tomcat server from *TOMCAT_HOME\bin\startup.bat* and locate the Web browser to *http://localhost:8080/Solr* (change the port if necessary). The admin page of Solr will be displayed.

What next?

Congratulations! You have understood the basics of Solr. You have learned about its syntax and different fields, how it makes search easy, the differences between Elastic Search and Solr, and how to deploy Tomcat with Solr. The next step is discovering how to connect Java (or any other language) with Solr, or, how to integrate Solr with your application. 

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